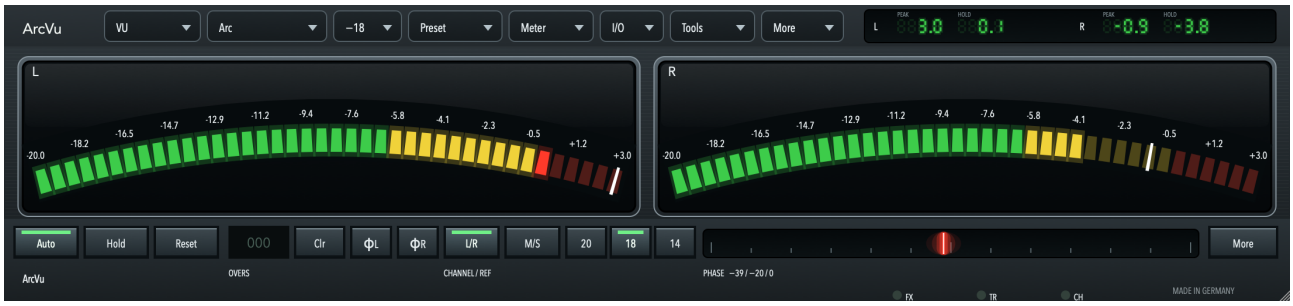
ArcVu 1.0.0 User Manual

ArcVu is a digital analogue meter by OpenGrid. It shows the level of your signal as a VU, RMS, K-System or peak programme (PPM) meter, with analogue-style needle dials (Classic skin) or lamp meters (Modern skin). The audio passes through ArcVu unchanged unless you switch on one of the monitoring tools.



ArcVu Arc face, Modern skin

Installing

ArcVu comes as VST3, Audio Unit, CLAP, LV2 and AAX plug-ins and a Standalone application (macOS), and as VST3, CLAP, LV2 and Standalone for Windows and Linux. The macOS installer puts the plug-ins in the system plug-in folders. In Pro Tools, ArcVu is an AAX plug-in signed for Pro Tools. The plug-in appears as **ArcVu** by **OpenGrid** in the Dynamics/Analyzer group of your host.

The interface



ArcVu Classic skin

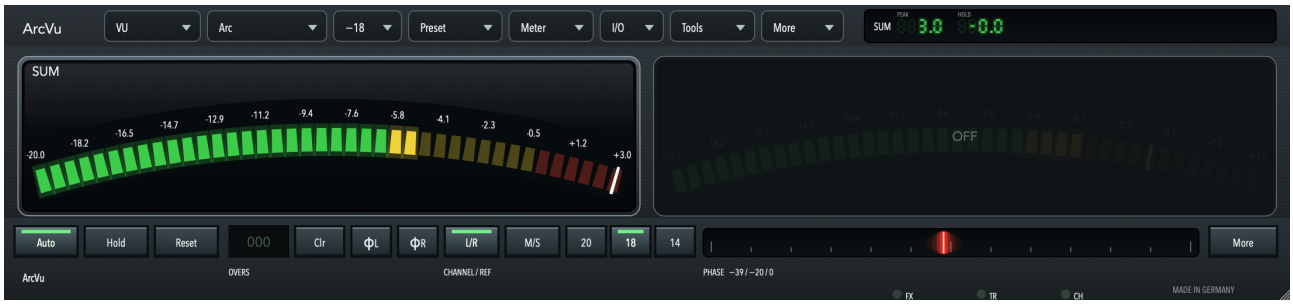
The header bar holds the menus. From left to right: the **meter type**, the **face**, the **reference level**, **Preset**, **Meter**, **I/O**, **Tools** and **More**. On the wide faces the digital PEAK/HOLD display sits at the right end of the header.

- **Faces.** *Arc* (two dials or arc meters side by side), *Horizontal* (two bars one above the other) and *Vertical* (two tall bars side by side with a shared scale).



Horizontal face

- **Skin** (More > Display). *Modern* draws 40-lamp meters. *Classic* draws analogue needle dials for all nine meter types.
- **Colour** (More > Display). Graphite/Ivory, Amber (a dark dial with orange markings in Classic) and Ice. Skin and colour are saved with your interface settings, not with presets or projects.
- **Size.** Drag the window corner to any size; the face keeps its proportions and ArcVu remembers the size. The 100%, 125% and 150% choice in the Display menu jumps to those sizes.
- **Digital readout.** PEAK and HOLD are shown as green seven-segment numbers. They stay on screen for the *Value hold* time (Meter menu, 5 seconds by default; set it from 0 to 30 seconds, or to -1 to hold until reset) so you can read them. The Values and Hold switches in the Display menu show or hide them.
- **Phase line.** A glowing line moves from out of phase (left) to mono (right). It turns red when the channels are mostly out of phase.
- **Keys row.** *Auto* makes the white peak lamp hold for the Peak hold time and then fall, *Hold* keeps it until you reset it, and with neither on (Follow) it follows the signal without holding. *Reset* clears peaks, the *OVERS* counter counts samples at or above full scale and *Clr* clears it, the polarity keys invert one channel for metering, *L/R* and *M/S* choose the channel mode, and *20*, *18* and *14* set the reference level in dBFS. The small lamps FX, TR and CH light when the processing tools, trims, or polarity/mute are active.
- **Single mode.** With Single on, the first meter shows the sum of both channels. The interface does not change: the second meter shows OFF.



Single mode: the second meter is off

Meter types

Type	What it shows
VU	The classic 300 ms ballistic (IEC 60268-17). 0 is the reference level.
RMS	RMS level in dBFS. With AES-17 on (the default) a full-scale sine reads 0 dBFS; turn it off for pure RMS (3 dB lower).

Type	What it shows
K-12, K-14, K-20	RMS relative to the K-System reference: 0 on the dial is -12, -14 or -20 dBFS RMS. The dial prints the reference under its title.
DIN, Nordic, BBC, EBU	Quasi-peak programme meters. Release times follow ITU-R BS.645-2 (DIN and Nordic 20 dB in 1.7 s, BBC 26 dB in 3 s, EBU 24 dB in 2.8 s). The EBU meter follows EBU Tech 3205-E (type IIb), has a slow mode (Preset menu), and shows TEST at 0 as on EBU scales.

The **reference level** (-20, -18 or -14 dBFS) is the level of a sine whose sample peak is at that value; the VU meter reads 0 there. The VU dial runs from -20 to +4 in whole numbers.



K-14 dial with its reference printed under the title



EBU peak programme meter in the Ice colour

Reading levels

The needle or lamp bar follows the meter's own ballistic. The white lamp (Modern) shows sample peak. A red lamp shows clipping. The digital PEAK number is the highest peak since it was last reset or since the Value hold time ran out, and HOLD is the highest meter reading held the same way. On VU and the PPM types both are in dB relative to the reference.

Menus

Preset. Step through factory presets (Broadcast VU, K-14, DIN peak, Podcast -14), switch between Factory and User presets, copy and paste settings, save and load preset files, save your own default

(Save default / Load default), save global settings (drag mode, drag sensitivity, refresh rate, tooltips), and turn on the EBU slow mode.

Meter. Peak lamp behaviour (Follow, Auto, Hold), Reset peak and Reset overs, the RMS window, AES-17, the VU model (Ideal 300 ms, Model 2, Custom), rise, fall, overshoot, peak hold, value hold, clip hold and clip threshold, the peak lamp threshold, the red zone, the comfort zone (K meters), analog RMS and the weighting filter.

I/O. Mid/side mode, trims for left, right, mid and side (with link), polarity, mute, the monitor selection (L, R, mid, side), Force mono and In place.

Tools. A dropdown of knobs. *High-pass* and *Low-pass* have a frequency and a slope (Off, 6, 12, 18 or 24 dB per octave). *Mono maker* narrows the side signal below its frequency by the amount you set. *Dynamic* is a band-limited downward gain with frequency, depth, type (high-pass, low-pass, narrow band, wide band), an on switch and a Listen switch to hear the band. Frequencies show at most three digits: 12.5 means 12.5 kHz, 380 means 380 Hz. The tools change what you monitor; with every tool off the audio is untouched.



Tools dropdown

More. Display (face, skin, colour, size, Single, Meters only, and the Values, Hold, Tools and Label switches), Preferences (drag sensitivity, meter refresh rate, bypass of hidden processors, tooltips, stay above windows).

Notes

- **Meters only** hides the lower panel. With *Bypass hidden processors* on, the tools that are no longer visible are bypassed.
- **Label.** Click the label area under the keys to type a name for the instance (up to 31 bytes of text). It is stored with the project.
- **Latency and CPU.** ArcVu adds no latency. The metering runs in the audio thread without memory allocation; drawing happens in the editor.
- **State.** All settings are saved with the project. Skin, colour and window size belong to the interface settings and are shared by all instances.

Parameter reference

Every host-automatable parameter, its range and its default value.

Meter type and reference

Parameter	Range / choices	Default
Meter type	VU, RMS, K-12, K-14, K-20, DIN, Nordic, BBC, EBU	VU
0 VU reference	-20 dBFS, -18 dBFS, -14 dBFS	-18 dBFS

Display

Parameter	Range / choices	Default
Face	Arc, Horizontal, Vertical	Arc
Single meter	Off / On	Off
GUI size	0 to 2	0
Meters only	Off / On	Off
Show values	Off / On	On
Show hold	Off / On	On
Show tools	Off / On	On
Show label	Off / On	On

Meter settings

Parameter	Range / choices	Default
Peak	Follow, Auto, Hold	Auto
RMS window	30.0 to 2000.0	300.0
AES-17	Off / On	On
VU model	0 to 2	0
Rise	50.0 to 1000.0	300.0
Fall	50.0 to 2000.0	300.0
Overshoot	0.2 to 8.0	1.0
Peak hold	-1.0 to 30.0	3.0
Value hold	-1.0 to 30.0	5.0
Clip hold	-1.0 to 30.0	2.0
Clip threshold	-12.0 to 0.0	0.0
Peak lamp threshold	-24.0 to 0.0	-6.0
Red zone	-12.0 to 12.0	0.0
Comfort zone	1.0 to 12.0	4.0
Analog RMS	Off / On	On
Weighting	0 to 5	0

I/O and channel tools

Parameter	Range / choices	Default
Mid side	Off / On	Off
Trim L	-24.0 to 24.0	0.0
Trim R	-24.0 to 24.0	0.0
Trim M	-24.0 to 24.0	0.0
Trim S	-24.0 to 24.0	0.0
Trim link	Off / On	Off
Polarity L	Off / On	Off
Polarity R	Off / On	Off
Mute L	Off / On	Off
Mute R	Off / On	Off

Parameter	Range / choices	Default
Monitor	LR, RL, L, R, Mid, Side	LR
Force mono	Off / On	Off
In place	Off / On	On

Processing tools

Parameter	Range / choices	Default
High-pass	20.0 to 500.0	80.0
High-pass slope	0 to 4	0
Low-pass	1000.0 to 20000.0	16000.0
Low-pass slope	0 to 4	0
Mono frequency	20.0 to 400.0	120.0
Mono amount	0.0 to 1.0	0.0
Dynamic on	Off / On	Off
Dynamic type	0 to 4	0
Dynamic frequency	80.0 to 12000.0	4000.0
Dynamic depth	0.0 to 1.0	0.0
Dynamic listen	Off / On	Off

Preferences

Parameter	Range / choices	Default
Drag sensitivity	0.25 to 4.0	1.0
Meter refresh	10 to 60	30
Bypass hidden processors	Off / On	Off
Tooltips	Off / On	On
Stay above windows	Off / On	Off

Standards and references

IEC 60268-17 (VU meter); ITU-R BS.645-2 (peak programme meters); EBU Tech 3205-E (type IIb PPM); AES-17 (RMS level); K-System by Bob Katz; BS 6840-17.

Licence and third-party software

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